

AutoVSR: Automatic Visual-to-Symbolic Reasoning for Symbolic Expression Generation from Circuit Schematic

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Verified executable IR + tool-augmented symbolic reasoning for reliable circuit analysis

Background

Symbolic expressions (e.g., transfer functions) are fundamental for analyzing and predicting circuit behavior.

- Symbolic expressions are crucial for characterizing and predicting circuit behavior.
- Deriving expressions directly from circuit schematics is difficult because it requires both visual understanding and correct multi-step symbolic reasoning.
- Existing end-to-end VLM approaches lack explicit intermediate verification, while symbolic toolchains such as SymPy and Lcapy require structured symbolic inputs rather than raw schematic images.

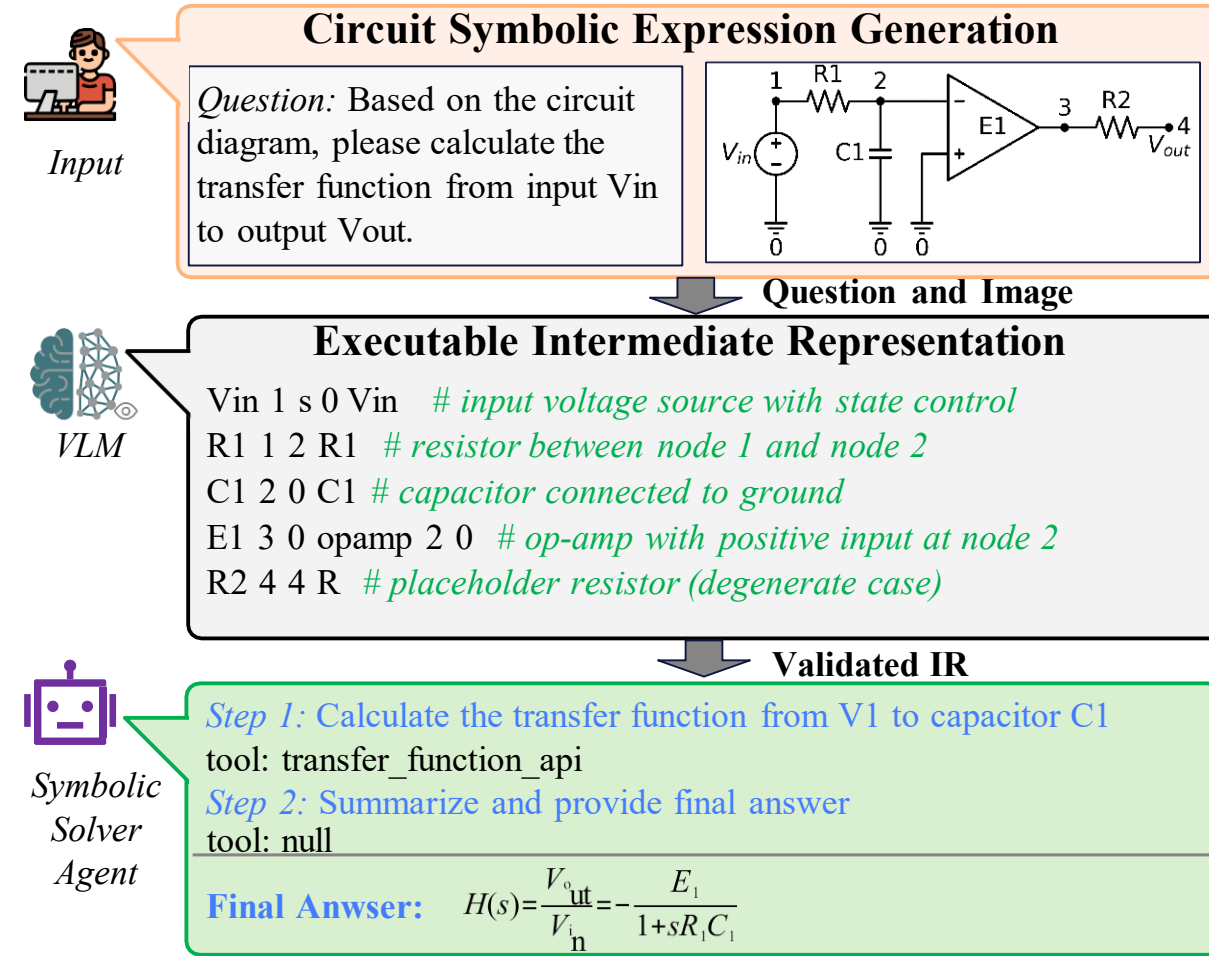
Challenges & Motivation

Challenge 1: Reliable visual-to-symbolic construction

Circuit schematics do not directly provide machine-readable device types, connectivity, or variable definitions; errors in reconstruction propagate to later reasoning.

Challenge 2: Correct multi-step symbolic derivation

Symbolic derivation is sensitive to sign conventions, equation setup, and algebraic errors; mistakes accumulate across steps.

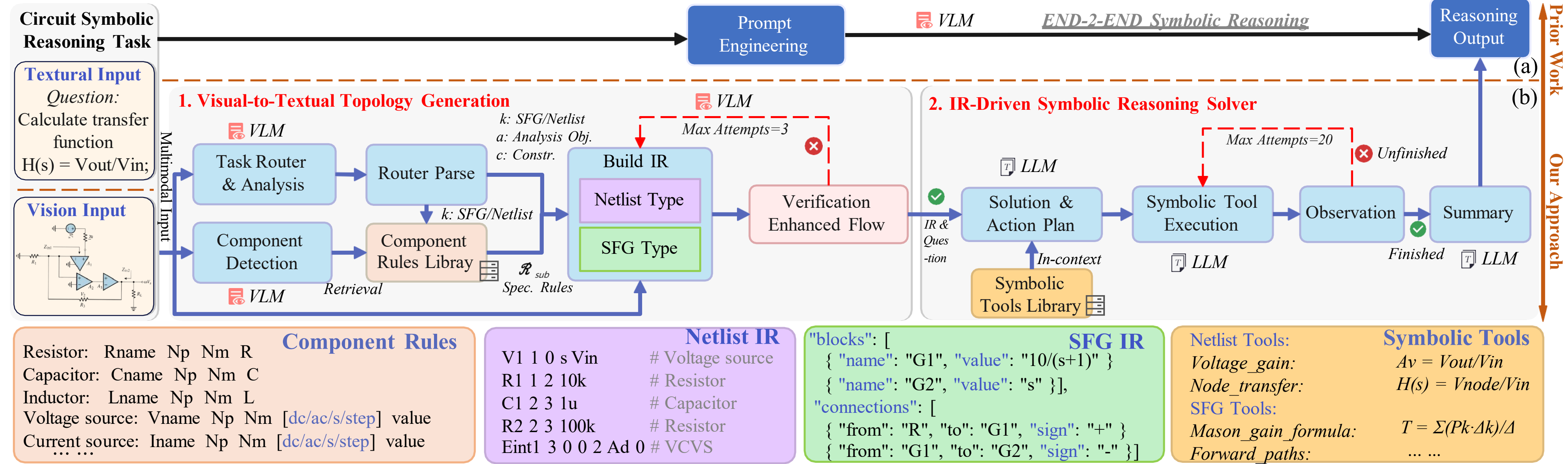


AutoVSR bridges schematic images and symbolic expressions through an executable, verifiable intermediate representation and a tool-based symbolic solver.

Contributions

- Propose AutoVSR, a visual-to-symbolic reasoning framework for symbolic expression generation from circuit schematics.
- Introduce an Executable IR with two forms: Netlist IR for schematic circuits and SFG IR for system-level signal-flow graphs.
- Design a verification-enhanced visual-to-textual topology generation stage with task routing, component rule retrieval, and iterative error-correction feedback.
- Build an IR-driven symbolic reasoning solver with plan-and-execute reasoning and a symbolic tool library, achieving strong gains in accuracy, efficiency, and interpretability.

Overview



Experiments

Table 1. Performance Comparison of AutoVSR and End-to-End VLMs on Transfer Function Generation Across Different Task Types.

Model Type	Method	Type 1		Type 2		Type 3		Type 4		Type 5		Overall	
		Acc. (%)	Imp. ↑	Acc. (%)	Imp. ↑	Acc. (%)	Imp. ↑	Acc. (%)	Imp. ↑	Acc. (%)	Imp. ↑	Acc. (%)	Imp. ↑
Closed Source	Gemini-3-Flash	37.95	+46.99	35.67	+53.51	23.06	+51.29	11.93	+72.73	3.95	+83.77	23.40	+59.45
	AutoVSR(Gemini-3-Flash)	84.94		89.18		74.35		84.66		87.72		82.85	
	GPT-5-mini	9.04	+83.13	9.36	+68.13	7.97	+23.50	13.64	+40.34	16.23	+14.91	10.54	+42.51
	AutoVSR(GPT-5-mini)	92.17		77.49		31.47		53.98		31.14		53.05	
	Qwen3-VL-Plus	18.07	+74.70	19.30	+74.85	13.36	+35.78	19.89	+50.00	15.79	+32.02	16.64	+51.38
	AutoVSR(Qwen3-VL-Plus)	92.77		94.15		49.14		69.89		47.81		68.02	
Open Source	Claude-Haiku-4.5	9.64	+64.46	10.53	+68.71	10.56	+18.10	9.09	+15.91	5.70	+14.04	9.67	+35.10
	AutoVSR(Claude-Haiku-4.5)	74.10		79.24		28.66		25.00		19.74		44.77	
	Qwen3-VL-32B-Instruct	15.06	+78.31	13.74	+75.73	7.11	+28.67	11.93	+58.52	17.98	+25.44	12.14	+49.63
	AutoVSR(Qwen3-VL-32B-Instruct)	93.37		89.47		35.78		70.45		43.42		61.77	
	Llama-4-17B-128E	10.84	+42.77	10.23	+45.33	6.47	+18.53	7.39	+28.41	13.16	+22.37	9.16	+30.01
	AutoVSR(Llama-4-17B-128E)	53.61		55.56		25.00		35.80		35.53		39.17	
Avg.	GLM-4.6V-Flash	3.61	+74.70	8.19	+65.20	6.25	+30.17	8.52	+34.09	10.09	+16.66	7.34	+42.51
	AutoVSR(GLM-4.6V-Flash)	78.31		73.39		36.42		42.61		26.75		49.85	
	Base Model	14.89	+66.43	15.29	+64.49	10.68	+29.44	11.77	+42.86	11.84	+29.89	12.70	+44.37
	AutoVSR	81.32		79.78		40.12		54.63		41.73		57.07	

Improvement range across models: +30.01 to +59.45 percentage points.

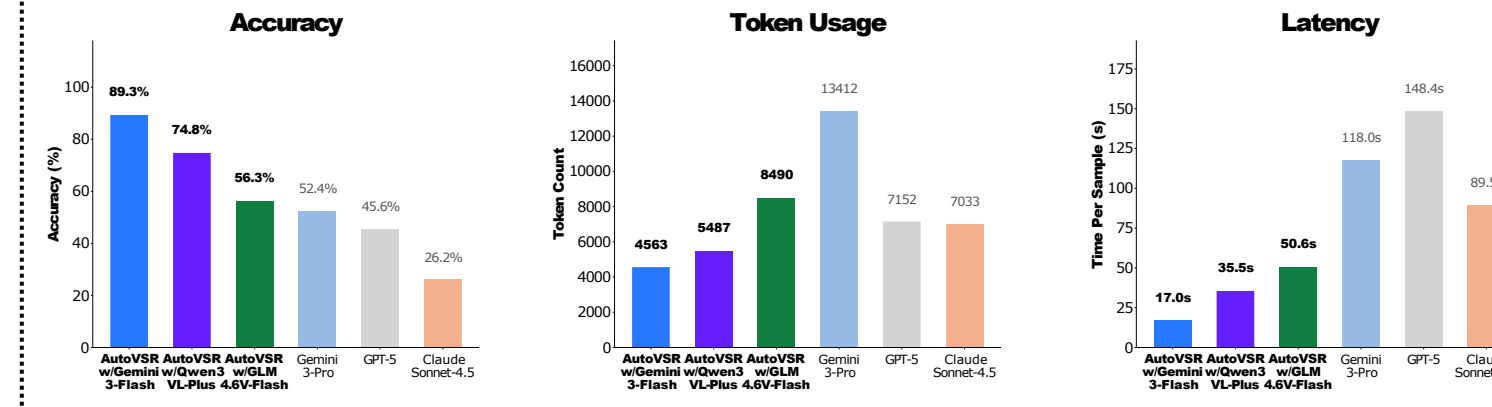
Benchmark: CircuitSense symbolic-expression-generation tasks, including transfer function generation and transient response expression generation across five circuit types.

Table 2. Performance Comparison between AutoVSR and CircuitSense.

Transfer Function Generation			
Model	CircuitSense	AutoVSR	Improvement(%)
Gemini-3-Flash	40.89	82.85	+ 41.96
Qwen3-VL-Plus	16.18	68.02	+ 51.84
GLM-4.6V-Flash	7.80	49.85	+ 42.05
Transfer Function Generation			
Gemini-3-Flash	3.10	85.84	+ 82.74
GLM-4.6V-Flash	1.02	63.25	+ 62.23

Deterministic symbolic tools and executable intermediate representations yield much stronger performance than circuit-specific end-to-end reasoning.

Fig 1. Practical Efficiency



AutoVSR with lightweight VLM backbones achieves higher accuracy with fewer tokens and lower inference runtime than expensive frontier VLMs.